

Midnight Wallet SDK Reference

1. Introduction

The Midnight Wallet SDK is a TypeScript library for managing all three token types on the Midnight blockchain: **shielded tokens** (zero-knowledge privacy-preserving), **unshielded tokens** (Night, the native token), and **dust** (fee tokens with time-dependent generation). The SDK provides HD key derivation, address encoding, transaction balancing and signing, chain synchronization, and ZK proof generation – all through a unified facade.

1.1 Architecture Overview

The SDK follows a layered architecture with 13 packages:

Layer 1: Foundation

- abstractions – Branded types (WalletSeed, WalletState, ProtocolVersion, ...)
- utilities – Domain-agnostic operations (SafeBigInt, BlobOps, ...)
- hd – HD wallet key derivation (BIP32/BIP39, CIP-1852)
- address-format – Bech32m address encoding (mn_ prefix)

Layer 2: Infrastructure Clients

- node-client – Polkadot RPC client for the Midnight node
- indexer-client – GraphQL client for chain indexer
- prover-client – HTTP client for ZK proof server

Layer 3: Capabilities

- capabilities – Shared services and pure operations
- /balancer – Coin selection and balance recipes
- /submission – Transaction submission lifecycle
- /pendingTransactions – Pending transaction tracking
- /proving – ZK proof generation services

Layer 4: Wallet Implementations

- shielded-wallet – Shielded (private) token wallet
- unshielded-wallet – Unshielded (Night) token wallet
- dust-wallet – Dust (fee) token wallet

Layer 5: Facade

- facade – Unified API combining all three wallets

1.2 Token Model

Midnight implements three distinct token types:

Token	Wallet	Privacy	Purpose
Custom shielded	ShieldedWallet	ZK-private	Arbitrary tokens with zero-knowledge proofs
Night (unshielded)	UnshieldedWallet	Public	Native token for transfers and staking
Dust	DustWallet	Public	Fee token with time-dependent generation

1.3 Variant/Runtime Pattern

Each wallet uses a **variant-based architecture** for protocol version compatibility. A variant is a specific implementation for a range of protocol versions (enabling hard-fork migration). The `WalletBuilder` registers variants, and the `Runtime` dispatches to the correct one based on the current protocol version.

1.4 Effect Library Integration

The SDK uses the [Effect](#) library extensively. Most services have two flavors:

- **Effect-based** (`*Effect`): Returns `Effect.Effect<T, E>` – composable, lazy, with typed errors. Used internally and for advanced integrations.
- **Promise-based**: Returns `Promise<T>` – standard async/await API for most consumers.

All errors extend `Data.TaggedError`, providing a `_tag` discriminant for pattern matching.

1.5 Evidence Summary

Tier	Passing	Failing	Errors	Total
tier1	95	0	0	95
tier3	0	0	1	1
Total	95	1		96

2. API Surface

@midnight-ntwrk/wallet-sdk-abstractions (2.0.0)

Symbol	Kind	Members
WalletSeed	namespace	—
WalletState	namespace	—
SerializedTransaction	namespace	—
ProtocolState	namespace	—
ProtocolVersion	namespace	—
NetworkId	namespace	—
SyncProgress	namespace	—

@midnight-ntwrk/wallet-sdk-address-format (3.1.0)

Symbol	Kind	Members
mainnet	const	—
NetworkId	type	—
FormatContext	type	—
Field	type	—
BLSScalar	const	—
ScaleBigInt	const	—
Bech32mSymbol	const	—
HasCodec	type	—
CodecTarget	type	—
MidnightBech32m	class	—
Bech32mCodec	class	—
ShieldedAddress	class	—
ShieldedEncryptionSecretKey	class	—
ShieldedCoinPublicKey	class	—
ShieldedEncryptionPublicKey	class	—
UnshieldedAddress	class	—
DustAddress	class	—

@midnight-ntwrk/wallet-sdk-capabilities (3.2.0)

Symbol	Kind	Members
InsufficientFundsError	class	—
BalanceRecipe	interface	—
CoinSelection	type	—
BalanceRecipeProps	type	—
getBalanceRecipe	const	—
createCounterOffer	const	—
chooseCoin	const	—
TransactionCostModel	interface	—
CounterOffer	class	—
TokenType	type	—
TokenValue	type	—
CoinRecipe	interface	—
Imbalance	type	—
Imbalances	type	—
PendingTransactionsService	type	—
IndexerClientConnection	type	—
DefaultPendingTransactionsServiceConfiguration	type	—
InitParams	type	—
PendingTransactionsServiceImpl	class	—
PendingTransactionsServiceEffect	type	—
PendingTransactionsServiceEffectImpl	class	—
PendingTransactions	namespace	—
ProvingError	class	—
ProvingServiceEffect	interface	—
ProvingService	interface	—
UnboundTransaction	type	—
fromProvingProviderEffect	const	—

Symbol	Kind	Members
fromProvingProvider	const	—
ServerProvingConfiguration	type	—
WasmProvingConfiguration	type	—
DefaultProvingConfiguration	type	—
makeServerProvingServiceEffect	const	—
makeWasmProvingServiceEffect	const	—
makeSimulatorProvingServiceEffect	const	—
makeDefaultProvingServiceEffect	const	—
makeDefaultProvingService	const	—
makeServerProvingService	const	—
makeWasmProvingService	const	—
makeSimulatorProvingService	const	—
SubmissionEvent	const	—
Finalized	type	—
Submitted	type	—
InBlock	type	—
SubmissionError	class	—
SubmitTransactionMethod	type	—
SubmitTransactionMethodEffect	type	—
SubmissionServiceEffect	interface	—
SubmissionService	interface	—
DefaultSubmissionConfiguration	type	—
makeDefaultSubmissionServiceEffect	const	—
makeDefaultSubmissionService	const	—
SimulatorSubmissionConfiguration	type	—
makeSimulatorSubmissionService	const	—

Sub-export: `./balancer`

Symbol	Kind	Members
InsufficientFundsError	class	—
BalanceRecipe	interface	—
CoinSelection	type	—
BalanceRecipeProps	type	—
getBalanceRecipe	const	—
createCounterOffer	const	—
chooseCoin	const	—
TransactionCostModel	interface	—
CounterOffer	class	—
TokenType	type	—
TokenValue	type	—
CoinRecipe	interface	—
Imbalance	type	—
Imbalances	type	—

Sub-export: `./submission`

Symbol	Kind	Members
SubmissionEvent	const	—
Finalized	type	—
Submitted	type	—
InBlock	type	—
SubmissionError	class	—
SubmitTransactionMethod	type	—
SubmitTransactionMethodEffect	type	—
SubmissionServiceEffect	interface	—
SubmissionService	interface	—
DefaultSubmissionConfiguration	type	—
makeDefaultSubmissionServiceEffect	const	—
makeDefaultSubmissionService	const	—
SimulatorSubmissionConfiguration	type	—
makeSimulatorSubmissionService	const	—

Sub-export: `./pendingTransactions`

Symbol	Kind	Members
PendingTransactionsService	type	—
IndexerClientConnection	type	—
DefaultPendingTransactionsServiceConfiguration	type	—
InitParams	type	—
PendingTransactionsServiceImpl	class	—
PendingTransactionsServiceEffect	type	—
PendingTransactionsServiceEffectImpl	class	—
PendingTransactions	namespace	—

Sub-export: `./proving`

Symbol	Kind	Members
ProvingError	class	—
ProvingServiceEffect	interface	—
ProvingService	interface	—
UnboundTransaction	type	—
fromProvingProviderEffect	const	—
fromProvingProvider	const	—
ServerProvingConfiguration	type	—
WasmProvingConfiguration	type	—
DefaultProvingConfiguration	type	—
makeServerProvingServiceEffect	const	—
makeWasmProvingServiceEffect	const	—
makeSimulatorProvingServiceEffect	const	—
makeDefaultProvingServiceEffect	const	—
makeDefaultProvingService	const	—
makeServerProvingService	const	—
makeWasmProvingService	const	—
makeSimulatorProvingService	const	—

@midnight-ntwrk/wallet-sdk-dust-wallet (3.0.0)

Symbol	Kind	Members
DustWalletCapabilities	type	—
DustWalletState	class	—
DustWalletAPI	type	—
DustWallet	type	—
DustWalletClass	interface	—
DefaultDustConfiguration	type	—

Sub-export: `./v1`

Symbol	Kind	Members
PublicKey	type	—
CoreWallet	type	—
DustWalletCapabilities	type	—
DustWalletState	class	—
DustWalletAPI	type	—
DustWallet	type	—
DustWalletClass	interface	—
DefaultDustConfiguration	type	—
Keys	namespace	—
Simulator	namespace	—
SyncService	namespace	—
Transacting	namespace	—
Context	type	—
AnyContext	type	—
V1Tag	const	—
DefaultRunningV1	type	—
RunningV1Variant	class	—
BaseV1Configuration	type	—
DefaultV1Configuration	type	—
DefaultV1Variant	type	—
V1Variant	type	—
DefaultV1Builder	type	—
V1Builder	class	—
Dust	type	—
DustWithNullifier	type	—
DustFullInfo	type	—
DustGenerationDetails	type	—

Symbol	Kind	Members
DustGenerationInfo	type	—
UtxoWithMeta	type	—
AnyTransaction	type	—
UnprovenDustSpend	type	—
NetworkId	type	—
TotalCostParameters	type	—
CoinsAndBalances	namespace	—

@midnight-ntwrk/wallet-sdk-facade (3.0.0)

Symbol	Kind	Members
FinalizedTransactionRecipe	type	–
UnboundTransactionRecipe	type	–
UnprovenTransactionRecipe	type	–
BalancingRecipe	type	–
TokenTransfer	interface	type , receiverAddress , amount
ShieldedTokenTransfer	type	–
UnshieldedTokenTransfer	type	–
CombinedTokenTransfer	type	–
CombinedSwapInputs	type	–
CombinedSwapOutputs	type	–
TransactionIdentifier	type	–
UtxoWithMeta	type	–
FacadeState	class	–
TermsAndConditions	type	–
FetchTermsAndConditionsConfiguration	type	–
DefaultConfiguration	type	–
InitParams	type	–
WalletFacade	class	–

@midnight-ntwrk/wallet-sdk-hd (3.0.1)

Symbol	Kind	Members
Roles	const	—
Role	type	—
HDWallet	class	—
AccountKey	class	—
RoleKey	class	—
CompositeRoleKey	class	—
mnemonicToWords	const	—
generateMnemonicWords	const	—
joinMnemonicWords	const	—
generateRandomSeed	const	—
validateMnemonic	const	—

@midnight-ntwrk/wallet-sdk-indexer-client (1.2.0)

Symbol	Kind	Members
Connect	const	—
Disconnect	const	—
BlockHash	const	—
FetchTermsAndConditions	const	—
TransactionStatus	const	—
DustLedgerEvents	const	—
ShieldedTransactions	const	—
UnshieldedTransactions	const	—
ZswapEvents	const	—
Maybe	type	—
InputMaybe	type	—
Exact	type	—
MakeOptional	type	—
MakeMaybe	type	—
MakeEmpty	type	—
Incremental	type	—
Scalars	type	—
Block	type	—
BlockOffset	type	—
CollapsedMerkleTree	type	—
CommitteeMember	type	—
ContractAction	type	—
ContractActionOffset	type	—
ContractBalance	type	—
ContractCall	type	—
ContractDeploy	type	—
ContractUpdate	type	—

Symbol	Kind	Members
DParameter	type	—
DParameterChange	type	—
DustGenerationDtimeUpdate	type	—
DustGenerationStatus	type	—
DustInitialUtxo	type	—
DustLedgerEvent	type	—
DustOutput	type	—
DustSpendProcessed	type	—
EpochInfo	type	—
EpochPerf	type	—
FirstValidEpoch	type	—
Mutation	type	—
MutationConnectArgs	type	—
MutationDisconnectArgs	type	—
ParamChange	type	—
PoolMetadata	type	—
PresenceEvent	type	—
Query	type	—
QueryBlockArgs	type	—
QueryCommitteeArgs	type	—
QueryContractActionArgs	type	—
QueryDustGenerationStatusArgs	type	—
QueryEpochPerformanceArgs	type	—
QueryEpochUtilizationArgs	type	—
QueryPoolMetadataArgs	type	—
QueryPoolMetadataListArgs	type	—
QueryRegisteredFirstValidEpochsArgs	type	—

Symbol	Kind	Members
QueryRegisteredPresenceArgs	type	—
QueryRegisteredSpoSeriesArgs	type	—
QueryRegisteredTotalsSeriesArgs	type	—
QuerySpoByPoolIdArgs	type	—
QuerySpoCompositeByPoolIdArgs	type	—
QuerySpoIdentitiesArgs	type	—
QuerySpoIdentityByPoolIdArgs	type	—
QuerySpoListArgs	type	—
QuerySpoPerformanceBySpoSkArgs	type	—
QuerySpoPerformanceLatestArgs	type	—
QueryStakeDistributionArgs	type	—
QueryStakePoolOperatorsArgs	type	—
QueryTransactionsArgs	type	—
RegisteredStat	type	—
RegisteredTotals	type	—
RegularTransaction	type	—
RelevantTransaction	type	—
Segment	type	—
ShieldedTransactionsEvent	type	—
ShieldedTransactionsProgress	type	—
Spo	type	—
SpoComposite	type	—
SpoIdentity	type	—
StakeShare	type	—
Subscription	type	—
SubscriptionBlocksArgs	type	—
SubscriptionContractActionsArgs	type	—

Symbol	Kind	Members
SubscriptionDustLedgerEventsArgs	type	—
SubscriptionShieldedTransactionsArgs	type	—
SubscriptionUnshieldedTransactionsArgs	type	—
SubscriptionZswapLedgerEventsArgs	type	—
SystemParameters	type	—
SystemTransaction	type	—
TermsAndConditions	type	—
TermsAndConditionsChange	type	—
Transaction	type	—
TransactionFees	type	—
TransactionOffset	type	—
TransactionResult	type	—
TransactionResultStatus	type	—
UnshieldedTransaction	type	—
UnshieldedTransactionsEvent	type	—
UnshieldedTransactionsProgress	type	—
UnshieldedUtxo	type	—
ZswapLedgerEvent	type	—
BlockHashQueryVariables	type	—
BlockHashQuery	type	—
ConnectMutationVariables	type	—
ConnectMutation	type	—
DisconnectMutationVariables	type	—
DisconnectMutation	type	—
FetchTermsAndConditionsQueryVariables	type	—
FetchTermsAndConditionsQuery	type	—
TransactionStatusQueryVariables	type	—

Symbol	Kind	Members
TransactionStatusQuery	type	—
DustLedgerEventsSubscriptionVariables	type	—
DustLedgerEventsSubscription	type	—
ShieldedTransactionsSubscriptionVariables	type	—
ShieldedTransactionsSubscription	type	—
UnshieldedTransactionsSubscriptionVariables	type	—
UnshieldedTransactionsSubscription	type	—
ZswapEventsSubscriptionVariables	type	—
ZswapEventsSubscription	type	—
BlockHashDocument	const	—
ConnectDocument	const	—
DisconnectDocument	const	—
FetchTermsAndConditionsDocument	const	—
TransactionStatusDocument	const	—
DustLedgerEventsDocument	const	—
ShieldedTransactionsDocument	const	—
UnshieldedTransactionsDocument	const	—
ZswapEventsDocument	const	—

Sub-export: `./effect`

Symbol	Kind	Members
Query	namespace	—
Subscription	namespace	—
QueryClient	class	—
HttpQueryClient	namespace	—
SubscriptionClient	class	—
WsSubscriptionClient	namespace	—
ConnectionHelper	namespace	—
QueryRunner	namespace	—

@midnight-ntwrk/wallet-sdk-node-client (1.1.0)

Symbol	Kind	Members
Config	type	—
makeConfig	const	—
DEFAULT_CONFIG	const	—
PolkadotNodeClient	class	—

Sub-export: ./effect

Symbol	Kind	Members
NodeClient	namespace	—
Config	type	—
DEFAULT_CONFIG	const	—
makeConfig	const	—
PolkadotNodeClient	class	—
SubmissionEvent	namespace	—
NodeClientError	namespace	—

Sub-export: ./testing

Symbol	Kind	Members
TestTransactions	namespace	—

@midnight-ntwrk/wallet-sdk-prover-client (1.2.0)

Symbol	Kind	Members
HttpProverClient	class	—

Sub-export: `./effect`

Symbol	Kind	Members
ProverClient	namespace	—
HttpProverClient	namespace	—
WasmProver	namespace	—

@midnight-ntwrk/wallet-sdk-runtime (1.0.2)

Symbol	Kind	Members
WalletBuilder	class	—
BuildArguments	type	—
FullConfiguration	type	—
Runtime	namespace	—

Sub-export: `./abstractions`

Symbol	Kind	Members
Variant	namespace	—
VariantBuilder	namespace	—
WalletLike	namespace	—
WalletRuntimeError	class	—
StateChange	namespace	—
VersionChangeType	namespace	—

@midnight-ntwrk/wallet-sdk-shielded (2.1.0)

Symbol	Kind	Members
ShieldedWalletCapabilities	type	—
UnboundTransaction	type	—
ShieldedWalletState	class	—
ShieldedWallet	type	—
ShieldedWalletClass	type	—
ShieldedWalletAPI	type	—
CustomizedShieldedWallet	type	—
DefaultShieldedConfiguration	type	—
CustomizedShieldedWalletClass	interface	—
CustomShieldedWallet	function	—

Sub-export: `./v1`

Symbol	Kind	Members
BaseV1Configuration	type	—
DefaultV1Configuration	type	—
V1Variant	type	—
AnyV1Variant	type	—
DefaultV1Variant	type	—
TransactionOf	type	—
AuxDataOf	type	—
SerializedStateOf	type	—
DefaultV1Builder	type	—
V1Builder	class	—
Sync	namespace	—
Transacting	namespace	—
TransactionHistory	namespace	—
Serialization	namespace	—
CoinsAndBalances	namespace	—
Keys	namespace	—
Context	type	—
AnyContext	type	—
V1Tag	const	—
DefaultRunningV1	type	—
RunningV1Variant	class	—
Simulator	namespace	—
WalletError	namespace	—
PublicKeys	type	—
CoinHashesMap	type	—
CoreWallet	type	—
TransactionOps	type	—

@midnight-ntwrk/wallet-sdk-unshielded-wallet (2.1.0)

Symbol	Kind	Members
UnshieldedWalletCapabilities	type	—
UnshieldedWalletState	class	—
UnshieldedWallet	type	—
DefaultUnshieldedConfiguration	type	—
UnshieldedWalletClass	type	—
UnshieldedWalletAPI	type	—
CustomizedUnshieldedWallet	type	—
CustomizedUnshieldedWalletClass	interface	—
CustomUnshieldedWallet	function	—
TransactionHistory	type	—
InMemoryTransactionHistoryStorage	class	—
NoOpTransactionHistoryStorage	class	—
TransactionHash	type	—
TransactionHistoryEntrySchema	const	—
TransactionHistoryEntry	type	—
TransactionHistoryStorage	interface	—
PublicKey	type	—
UnshieldedKeystore	interface	—
createKeystore	const	—

Sub-export: `./v1`

Symbol	Kind	Members
BaseV1Configuration	type	—
DefaultV1Configuration	type	—
V1Variant	type	—
AnyV1Variant	type	—
DefaultV1Variant	type	—
TransactionOf	type	—
SerializedStateOf	type	—
DefaultV1Builder	type	—
V1Builder	class	—
Sync	namespace	—
SyncProgress	namespace	—
Transacting	namespace	—
TransactionHistory	namespace	—
Serialization	namespace	—
CoinsAndBalances	namespace	—
Keys	namespace	—
Context	type	—
AnyContext	type	—
V1Tag	const	—
DefaultRunningV1	type	—
RunningV1Variant	class	—
Simulator	namespace	—
WalletError	namespace	—
CoreWallet	type	—
UnboundTransaction	type	—
IntentOf	type	—
TransactionOps	type	—

Symbol	Kind	Members
TransactionHistoryStorage	namespace	—
UnshieldedState	namespace	—

@midnight-ntwrk/wallet-sdk-utilities (1.1.0)

Symbol	Kind	Members
ArrayOps	namespace	—
BlobOps	namespace	—
DateOps	namespace	—
EitherOps	namespace	—
Fluent	namespace	—
HList	namespace	—
LedgerOps	namespace	—
ObservableOps	namespace	—
Poly	namespace	—
RecordOps	namespace	—
SafeBigInt	namespace	—

Sub-export: ./networking

Symbol	Kind	Members
HttpURL	namespace	—
WsURL	namespace	—
InvalidProtocolSchemeError	class	—
FailedToDeriveWebSocketUrlError	class	—
URLError	type	—
ClientError	class	—
ServerError	class	—

Sub-export: ./types

Symbol	Kind	Members
CanAssign	type	—
Expect	type	—
ItemType	type	—
Equal	type	—

Sub-export: `./testing`

Symbol	Kind	Members
TestContainers	namespace	—
getRepositoryRoot	function	—
getComposeDirectory	function	—
BuildTestEnvironmentVariablesOptions	interface	—
buildTestEnvironmentVariables	function	—

Total: 471 symbols, 3 members across 13 packages

3. Foundation Layer

3.1 Abstractions (@midnight-ntwrk/wallet-sdk-abstractions)

Branded types that form the domain vocabulary of the wallet SDK. Each type is a nominal brand using Effect's `Brand` module, preventing accidental use of raw primitives where a branded type is expected.

Export	Kind	Description
<code>WalletSeed</code>	namespace	Branded <code>Uint8Array</code> for BIP32 seeds. Has <code>fromString(hex)</code> , <code>is(value)</code> predicate.
<code>WalletState</code>	namespace	Branded <code>string</code> for serialized wallet state. Has <code>is(value)</code> predicate.
<code>SerializedTransaction</code>	namespace	Branded <code>Uint8Array</code> for serialized transactions. Has <code>of(bytes)</code> and <code>from(serializable)</code> helpers.
<code>ProtocolVersion</code>	namespace	Branded <code>bigint</code> for protocol versions. Has <code>makeRange</code> , <code>withinRange</code> , <code>is</code> , <code>MinSupportedVersion (0)</code> , <code>MaxSupportedVersion</code> .
<code>NetworkId</code>	namespace	Well-known network identifiers: <code>MainNet</code> , <code>TestNet</code> , <code>DevNet</code> , <code>QaNet</code> , <code>Undeployed</code> , <code>Preview</code> , <code>PreProd</code> . Accepts any string.
<code>SyncProgress</code>	namespace	Tracks wallet sync state. <code>createSyncProgress(params)</code> creates instances with <code>isStrictlyComplete()</code> and <code>isCompleteWithin(maxGap)</code> methods.
<code>ProtocolState</code>	namespace	Wraps wallet state with protocol version metadata.

Test	Status	Detail
abstractions/WalletSeed-from-bytes	PASS	—
abstractions/WalletSeed-from-string	PASS	—
abstractions/WalletSeed-is-predicate	PASS	—
abstractions/WalletState-from-string	PASS	—
abstractions/SerializedTransaction-of	PASS	—
abstractions/ProtocolVersion-construction	PASS	—
abstractions/ProtocolVersion-min-max	PASS	—
abstractions/ProtocolVersion-range-within	PASS	—
abstractions/ProtocolVersion-is-predicate	PASS	—
abstractions/NetworkId-well-known	PASS	—
abstractions/SyncProgress-create	PASS	—
abstractions/SyncProgress-strictly-complete	PASS	—
abstractions/SyncProgress-not-complete	PASS	—

3.2 Utilities (@midnight-ntwrk/wallet-sdk-utilities)

Domain-agnostic operations used across the SDK. Each utility is exported as a namespace module containing related functions.

Export	Kind	Description
SafeBigInt	namespace	Safe conversions between bigint and other types.
ArrayOps	namespace	Array utility functions (filter, map, group).
BlobOps	namespace	Binary/blob operations (concat, slice, compare).
DateOps	namespace	Date arithmetic (e.g., addSeconds).
EitherOps	namespace	Effect Either utilities.
Fluent	namespace	Fluent API builder helpers.
HList	namespace	Heterogeneous list types (used by WalletBuilder).
LedgerOps	namespace	Ledger-specific type operations.
ObservableOps	namespace	RxJS Observable utilities.
Poly	namespace	Polymorphic function utilities (variant dispatch).
RecordOps	namespace	Record/object utilities.

Sub-export `./networking`: URL validation types and HTTP error classes.

Export	Kind	Description
HttpURL	namespace	Validated HTTP URL type.
WsURL	namespace	Validated WebSocket URL type.
InvalidProtocolSchemeError	class	Tagged error (<code>_tag: 'InvalidProtocolSchemeError'</code>) for invalid URL schemes.
FailedToDeriveWebSocketUrlError	class	Tagged error for WebSocket URL derivation failures.
ClientError	class	Tagged error (<code>_tag: 'ClientError'</code>) for HTTP 4xx responses.
ServerError	class	Tagged error (<code>_tag: 'ServerError'</code>) for HTTP 5xx responses.

Sub-export `./types`: Compile-time type utilities (`Expect`, `Equal`, `CanAssign`, `ItemType`).

Test	Status	Detail
utilities/SafeBigInt-namespace-exists	PASS	—
utilities/DateOps-namespace-exists	PASS	—
utilities/ArrayOps-namespace-exists	PASS	—
utilities/BlobOps-namespace-exists	PASS	—
utilities/InvalidProtocolSchemeError-construction	PASS	—
utilities/FailedToDeriveWebSocketUrlError-construction	PASS	—
utilities/ClientError-construction	PASS	—
utilities/ServerError-construction	PASS	—

3.3 HD Wallet (@midnight-ntwrk/wallet-sdk-hd)

Hierarchical deterministic key derivation following BIP32/BIP39 with Midnight's derivation path: `m/44'/2400'/{account}'/{role}/{index}` .

Coin type `2400` is Midnight's registered BIP44 coin type. Five roles partition keys by purpose:

Role	Index	Purpose
NightExternal	0	Receiving Night tokens
NightInternal	1	Change addresses for Night
Dust	2	Dust generation addresses
Zswap	3	Zero-knowledge swap keys
Metadata	4	Metadata encryption keys

Export	Kind	Description
HDWallet	class	HD wallet root. <code>fromSeed(seed)</code> returns <code>{ type: 'seedOk', hdWallet }</code> or <code>{ type: 'seedError', error }</code> . Call <code>selectAccount(n)</code> to get an <code>AccountKey</code> .
AccountKey	class	Account-level key. <code>selectRole(role)</code> gets a <code>RoleKey</code> ; <code>selectRoles([...])</code> gets a <code>CompositeRoleKey</code> for batch derivation.
RoleKey	class	Role-level key. <code>deriveKeyAt(index)</code> returns <code>{ type: 'keyDerived', key: Uint8Array }</code> or <code>{ type: 'keyOutOfBounds' }</code> .
CompositeRoleKey	class	Multi-role key. <code>deriveKeysAt(index)</code> derives keys for all specified roles simultaneously.
Roles	const	Enum object: <code>{ NightExternal: 0, NightInternal: 1, Dust: 2, Zswap: 3, Metadata: 4 }</code> .
<code>generateMnemonicWords</code>	function	Generates a 24-word BIP39 mnemonic (default strength=256).
<code>joinMnemonicWords</code>	function	Joins word array into space-separated mnemonic string.
<code>mnemonicToWords</code>	function	Splits mnemonic string into word array.
<code>validateMnemonic</code>	function	Validates a BIP39 mnemonic against the English wordlist.
<code>generateRandomSeed</code>	function	Generates a cryptographically random 32-byte seed.

Key derivation is **deterministic**: the same seed always produces the same keys. Different roles produce different keys. Different accounts produce different keys.

Test	Status	Detail
hd/mnemonic-generates-24-words	PASS	—
hd/mnemonic-validates-roundtrip	PASS	—
hd/invalid-mnemonic-rejected	PASS	—
hd/random-seed-length	PASS	—
hd/wallet-from-seed	PASS	—
hd/roles-values	PASS	—
hd/derive-key-at-index-0	PASS	—
hd/different-roles-different-keys	PASS	—
hd/different-accounts-different-keys	PASS	—
hd/composite-role-keys	PASS	—
hd/deterministic-derivation	PASS	—

3.4 Address Format (@midnight-ntwrk/wallet-sdk-address-format)

Bech32m address encoding with the `mn` prefix. Addresses encode a type identifier, network identifier, and payload into a single checksummed string.

Address Type	Bech32m Type	Payload	Purpose
ShieldedAddress	shield-addr	Coin public key + encryption public key	Receiving shielded tokens
UnshieldedAddress	addr	32-byte public key	Receiving Night tokens
DustAddress	dust	BLS scalar (bigint < modulus)	Dust generation registration
ShieldedCoinPublicKey	shield-cpk	32 bytes	Proving coin ownership
ShieldedEncryptionPublicKey	shield-epk	32 bytes	Encrypting messages
ShieldedEncryptionSecretKey	shield-esk	32 bytes	Decrypting messages

Export	Kind	Description
<code>MidnightBech32m</code>	class	Encodes/decodes addresses. <code>encode(networkId, item)</code> produces a Bech32m string; <code>parse(str) + decode(Type, networkId)</code> reverses it.
<code>Bech32mCodec</code>	class	Generic codec for custom Bech32m types.
<code>BLSScalar</code>	const	BLS12-381 scalar field: <code>{ bytes: 32, modulus: 0x73eda753...01 }</code> .
<code>ScaleBigInt</code>	const	SCALE codec for bigint encoding/decoding. Roundtrips correctly for all non-negative values.
<code>ShieldedAddress</code>	class	Composed of <code>coinPublicKey</code> + <code>encryptionPublicKey</code> .
<code>ShieldedCoinPublicKey</code>	class	32-byte key with <code>toHexString()</code> , <code>equals()</code> , <code>fromHexString()</code> . Accepts Buffer, not Uint8Array.
<code>ShieldedEncryptionPublicKey</code>	class	32-byte key with <code>toHexString()</code> , <code>equals()</code> . Accepts Buffer.
<code>UnshieldedAddress</code>	class	32-byte address with <code>hexString</code> getter and <code>equals()</code> . Accepts Buffer.
<code>DustAddress</code>	class	Stores a <code>bigint</code> validated against <code>BLSScalar.modulus</code> . Values <code>>= modulus</code> are rejected.
<code>mainnet</code>	symbol	Unique symbol for mainnet network ID (not a string).

Important: Address constructors accept `Buffer`, not `Uint8Array`. This is a platform-specific choice aligned with Node.js conventions.

Test	Status	Detail
address-format/BLSScalar-is-32-bytes	PASS	—
address-format/ScaleBigInt-encode-decode-roundtrip	PASS	—
address-format/ScaleBigInt-zero	PASS	—
address-format/ScaleBigInt-large-value	PASS	—
address-format/ShieldedCoinPublicKey-construction	PASS	—
address-format/ShieldedCoinPublicKey-equality	PASS	—
address-format/ShieldedEncryptionPublicKey-construction	PASS	—
address-format/ShieldedAddress-composition	PASS	—
address-format/UnshieldedAddress-hex-roundtrip	PASS	—
address-format/UnshieldedAddress-equality	PASS	—
address-format/DustAddress-from-bigint	PASS	—
address-format/DustAddress-rejects-over-modulus	PASS	—
address-format/DustAddress-validates-modulus	PASS	—
address-format/MidnightBech32m-encode-shielded	PASS	—
address-format/MidnightBech32m-encode-unshielded	PASS	—
address-format/MidnightBech32m-roundtrip	PASS	—

4. Infrastructure Clients

4.1 Node Client (@midnight-ntwrk/wallet-sdk-node-client)

Polkadot RPC client for submitting transactions to the Midnight node. Uses the @polkadot/api library over WebSocket.

Export	Kind	Description
PolkadotNodeClient	class	Main client. <code>init(config)</code> connects to the node. <code>sendMidnightTransaction(tx)</code> returns an <code>Observable<SubmissionEvent></code> tracking the submission lifecycle. <code>close()</code> disconnects.
Config	type	<code>{ nodeURL: URL, reconnectionTimeout: Duration, reconnectionDelay: Duration } .</code>
makeConfig	function	Creates a <code>Config</code> from partial input. Only <code>nodeURL</code> is required; timeouts have defaults.
DEFAULT_CONFIG	const	Default reconnection settings (infinite timeout, 1s delay). Does not include <code>nodeURL</code> .

The submission event stream emits `Submitted` , `InBlock` , and `Finalized` events as the transaction progresses through the block pipeline.

No evidence collected yet.

4.2 Indexer Client (@midnight-ntwrk/wallet-sdk-indexer-client)

GraphQL client for querying the chain indexer. The indexer tracks wallet-relevant events (UTxO creation/spending, shielded transactions, dust generation).

Queries:

Query	Parameters	Purpose
Connect	viewingKey: ViewingKey	Register a viewing key to start a session
Disconnect	sessionId: HexEncoded	End an indexer session
BlockHash	offset?: BlockOffset	Get block height, hash, ledger params, timestamp
TransactionStatus	transactionId: HexEncoded	Check transaction result and segment status

Subscriptions:

Subscription	Purpose
DustLedgerEvents	Stream dust ledger events for sync
ShieldedTransactions	Stream shielded transactions with Merkle tree updates
UnshieldedTransactions	Stream unshielded UTxO events
ZswapEvents	Stream Zswap ledger events

Effect sub-export (./effect):

Export	Kind	Description
QueryClient	tag	Effect Context Tag for GraphQL query execution.
SubscriptionClient	tag	Effect Context Tag for GraphQL subscription streaming.
HttpQueryClient	namespace	layer(config) creates an Effect Layer for HTTP queries.
WsSubscriptionClient	namespace	layer(config) creates an Effect Layer for WebSocket subscriptions.
ConnectionHelper	namespace	deriveWebSocketUrl(url) converts HTTP to WS URL.
QueryRunner	namespace	runPromise(query, vars, config) executes a query as a plain Promise.

Note: This package also exports ~120 auto-generated GraphQL types from the indexer schema (see Appendix A).

Test	Status	Detail
indexer-client/Connect-query-exists	PASS	—
indexer-client/Disconnect-query-exists	PASS	—
indexer-client/BlockHash-query-exists	PASS	—
indexer-client/TransactionStatus-query-exists	PASS	—
indexer-client/QueryClient-is-context-tag	PASS	—
indexer-client/SubscriptionClient-is-context-tag	PASS	—
indexer-client/ConnectionHelper-deriveWebSocketUrl	PASS	—
indexer-client/QueryRunner-runPromise	PASS	—

4.3 Prover Client (@midnight-ntwrk/wallet-sdk-prover-client)

HTTP client for the ZK proof generation server. The proof server converts `UnprovenTransaction` objects into `Transaction` objects with valid ZK proofs.

Export	Kind	Description
<code>HttpProverClient</code>	class	Wraps an HTTP connection. Constructor takes <code>{ url: string }</code> . <code>proveTransaction(tx, costModel?)</code> sends the transaction for proving and returns the proven version.

The Effect sub-export (`./effect`) provides `ProverClient` (Context Tag), `HttpProverClient` layer, and `SimulatorProverClient` for testing.

No evidence collected yet.

5. Capabilities

Shared capability implementations decomposed into pure state operations (capabilities) and side-effecting services. Every service has both an Effect-based variant (`*Effect`) and a plain Promise-based variant.

5.1 Balancer (`@midnight-ntwrk/wallet-sdk-capabilities/balancer`)

Coin selection and transaction balancing. Computes which UTXOs to consume and what change outputs to create to satisfy a set of imbalances.

Export	Kind	Description
<code>getBalanceRecipe</code>	function	Core balancing algorithm. Takes coins, imbalances, cost model, and coin selection strategy; returns a <code>BalanceRecipe</code> with inputs and outputs.
<code>createCounterOffer</code>	function	Creates a <code>CounterOffer</code> for multi-party transaction negotiation.
<code>chooseCoin</code>	function	Default coin selection: picks the smallest-value coin of the requested type.
<code>BalanceRecipe</code>	interface	<code>{ inputs: TInput[], outputs: TOutput[] }</code> – the result of balancing.
<code>CoinRecipe</code>	interface	<code>{ type: TokenType, value: TokenValue }</code> – a coin's type and value.
<code>TransactionCostModel</code>	interface	<code>{ inputFeeOverhead: bigint, outputFeeOverhead: bigint }</code> – per-input/output fee constants.
<code>CounterOffer</code>	class	Mutable accumulator for multi-step balancing with <code>addInput</code> / <code>addOutput</code> .
<code>Imbalances</code>	type/const	<code>Map<TokenType, TokenValue></code> with companion methods: <code>empty()</code> , <code>fromEntry()</code> , <code>merge()</code> , <code>getValue()</code> , <code>typeSet()</code> .
<code>InsufficientFundsError</code>	class	Thrown when balancing cannot satisfy an imbalance. Has <code>tokenType</code> property.

No evidence collected yet.

5.2 Submission ([@midnight-ntwrk/wallet-sdk-capabilities/submission](https://github.com/midnight-ntwrk/wallet-sdk-capabilities/blob/main/submission))

Transaction submission lifecycle: submit → in-block → finalized.

Export	Kind	Description
<code>SubmissionService</code>	interface	<code>prove(tx): Promise<T></code> – submits a transaction.
<code>SubmissionServiceEffect</code>	interface	Effect-based variant.
<code>SubmissionEvent</code>	const/type	Tagged union of <code>Submitted</code> , <code>InBlock</code> , <code>Finalized</code> events (from <code>Data.taggedEnum</code>).
<code>SubmissionError</code>	class	Tagged error (<code>_tag: 'SubmissionError'</code>) with { <code>message</code> , <code>cause?</code> } .
<code>makeDefaultSubmissionService</code>	function	Creates a submission service from a <code>DefaultSubmissionConfiguration</code> ({ <code>relayURL: URL</code> }).
<code>makeSimulatorSubmissionService</code>	function	Creates a mock submission service for testing.

No evidence collected yet.

5.3 Pending Transactions (`@midnight-ntwrk/wallet-sdk-capabilities/pendingTransactions`)

Tracks transactions that have been submitted but not yet finalized. Monitors the indexer for confirmation and expiry.

Export	Kind	Description
<code>PendingTransactionsService</code>	type	Service interface for tracking pending transactions.
<code>PendingTransactionsServiceImpl</code>	class	Promise-based implementation.
<code>PendingTransactions</code>	namespace	Contains <code>TransactionTrait</code> interface for transaction identity operations (serialize, deserialize, ID extraction, TTL checking).

No evidence collected yet.

5.4 Proving (`@midnight-ntwrk/wallet-sdk-capabilities/proving`)

ZK proof generation services. Multiple backends for different environments.

Export	Kind	Description
<code>ProvingService</code>	interface	<code>prove(tx: UnprovenTransaction): Promise<T></code> – generates a ZK proof for a transaction.
<code>ProvingServiceEffect</code>	interface	Effect-based variant.
<code>ProvingError</code>	class	Tagged error (<code>_tag: 'Wallet.Proving'</code>) with required <code>cause: Error</code> .
<code>makeDefaultProvingService</code>	function	Creates a server-backed proving service (requires { <code>provingServerUrl: URL</code> }).
<code>makeServerProvingService</code>	function	Explicit server-backed proving service.
<code>makeWasmProvingService</code>	function	In-browser WASM proving (optional <code>keyMaterialProvider</code>).
<code>makeSimulatorProvingService</code>	function	Mock proving for testing – returns <code>ProofErasedTransaction</code> (no real proof).
<code>UnboundTransaction</code>	type	A transaction with a real proof but pre-binding.

No evidence collected yet.

6. Wallet Implementations

All three wallets follow the same structural pattern:

1. A **factory function** (`ShieldedWallet(config)`) returns a wallet class
2. The class has `startWithSeed` / `startWithSecretKeys` / `restore` static methods
3. Each instance exposes `state: Observable<WalletState>`, lifecycle methods (`start`, `stop`), and token-specific operations
4. The `WalletState` class wraps capabilities (serialization, coins/balances, keys, transaction history) and provides convenient getters

6.1 Shielded Wallet (`@midnight-ntwrk/wallet-sdk-shielded`)

Privacy-preserving wallet for custom shielded tokens. Uses zero-knowledge proofs for all operations – balances and transaction details are hidden from observers.

Export	Kind	Description
<code>ShieldedWallet</code>	function	Factory: <code>ShieldedWallet(config)</code> → <code>ShieldedWalletClass</code> .
<code>ShieldedWalletClass</code>	type	Class with <code>startWithSeed(seed)</code> , <code>startWithSecretKeys(keys)</code> , <code>restore(state)</code> .
<code>ShieldedWalletState</code>	class	State with getters: <code>balances</code> (record of token balances), <code>totalCoins</code> , <code>availableCoins</code> , <code>pendingCoins</code> , <code>coinPublicKey</code> , <code>encryptionPublicKey</code> , <code>address</code> , <code>progress</code> .
<code>ShieldedWalletAPI</code>	type	Instance API: <code>state</code> , <code>start(secretKeys)</code> , <code>balanceTransaction(keys, tx)</code> , <code>transferTransaction(keys, outputs)</code> , <code>initSwap(keys, inputs, outputs)</code> , <code>getAddress()</code> , <code>revertTransaction(tx)</code> , <code>stop()</code> .
<code>DefaultShieldedConfiguration</code>	type	Default configuration type (aliases <code>DefaultV1Configuration</code>).

The shielded wallet requires `ZswapSecretKeys` for most operations, ensuring that only the key holder can view or modify the wallet state.

No evidence collected yet.

6.2 Unshielded Wallet (`@midnight-ntwrk/wallet-sdk-unshielded-wallet`)

Wallet for Night (the native token) and other unshielded tokens. Operations are publicly visible on the ledger.

Export	Kind	Description
<code>UnshieldedWallet</code>	function	Factory: <code>UnshieldedWallet(config)</code> → <code>UnshieldedWalletClass</code> .
<code>UnshieldedWalletClass</code>	type	Class with <code>startWithPublicKey(pk)</code> , <code>restore(state)</code> .
<code>UnshieldedWalletState</code>	class	State with getters: <code>balances</code> , <code>totalCoins</code> , <code>availableCoins</code> , <code>pendingCoins</code> , <code>address</code> , <code>progress</code> , <code>transactionHistory</code> .
<code>UnshieldedWalletAPI</code>	type	Instance API with three balance methods: <code>balanceFinalizedTransaction</code> , <code>balanceUnboundTransaction</code> , <code>balanceUnprovenTransaction</code> . Also <code>transferTransaction(outputs, ttl)</code> , <code>initSwap(inputs, outputs, ttl)</code> , signing methods.
<code>createKeystore</code>	function	<code>createKeystore(secretKey: Uint8Array, networkId): UnshieldedKeystore</code> – creates a keystore from a derived secret key.
<code>PublicKey</code>	type/const	<code>{ publicKey, addressHex, address }</code> with <code>fromKeystore(keystore)</code> factory.
<code>TransactionHistoryStorage</code>	interface	<code>create</code> , <code>delete</code> , <code>getAll</code> , <code>get</code> for transaction history persistence.

Unlike the shielded wallet, the unshielded wallet uses public-key based authentication and provides separate balancing methods for each transaction stage.

No evidence collected yet.

6.3 Dust Wallet (@midnight-ntwrk/wallet-sdk-dust-wallet)

Fee management wallet. Dust is generated over time from registered Night UTxOs. Each dust coin has generation parameters:

- **backingNight**: The Night UTxO backing this dust generation
- **rate**: Dust generation rate (dust per unit time)
- **maxCap**: Maximum dust that can accumulate
- **ctime**: Creation time of the dust generation

The balance at any given time is computed from these parameters, making it **time-dependent**: `balance(date)` returns the current dust amount as a `bigint` (in SPECK, the smallest dust unit).

Export	Kind	Description
DustWallet	function	Factory: <code>DustWallet(config) → DustWalletClass</code> .
DustWalletClass	type	Class with <code>startWithSeed(seed, dustParams)</code> , <code>startWithSecretKey(key, dustParams)</code> , <code>restore(state)</code> .
DustWalletState	class	State with getters: <code>address</code> , <code>totalCoins</code> , <code>availableCoins</code> , <code>pendingCoins</code> , <code>progress</code> , and methods: <code>balance(time)</code> (returns <code>bigint</code>), <code>availableCoinsWithFullInfo(time)</code> , <code>estimateDustGeneration(nightUtxos, time)</code> .
DustWalletAPI	type	Instance API: <code>state</code> , <code>start(secretKey)</code> , <code>createDustGenerationTransaction(...)</code> , <code>calculateFee(txs)</code> , <code>estimateFee(...)</code> , <code>balanceTransactions(...)</code> , <code>getAddress()</code> , <code>stop()</code> .
DefaultDustConfiguration	type	Default configuration type.

No evidence collected yet.

7. Facade (@midnight-ntwrk/wallet-sdk-facade)

The unified entry point. Orchestrates all three wallet types for transaction balancing – running unshielded, shielded, and dust/fee balancing in sequence, then merging results.

Export	Kind	Description
WalletFacade	class	Main facade. <code>init(params)</code> creates an instance. Exposes <code>state()</code> (Observable), <code>submitTransaction</code> , three <code>balance*</code> methods, <code>transferTransaction</code> , <code>initSwap</code> , dust registration/deregistration, and lifecycle methods.
FacadeState	class	Combined state: <code>shielded</code> , <code>unshielded</code> , <code>dust</code> sub-states plus <code>pending</code> . <code>isSynced</code> getter checks all three are synced.
BalancingRecipe	type/const	Union of <code>FinalizedTransactionRecipe</code> , <code>UnboundTransactionRecipe</code> , <code>UnprovenTransactionRecipe</code> . <code>isRecipe(value)</code> type guard and <code>getTransactions(recipe)</code> extractor.
TokenTransfer	type	<code>{ type: RawTokenType, receiverAddress: AddressType, amount: bigint }</code> .
ShieldedTokenTransfer	type	Transfer targeting shielded addresses.
UnshieldedTokenTransfer	type	Transfer targeting unshielded addresses.
CombinedTokenTransfer	type	Union of shielded and unshielded transfers.
CombinedSwapInputs	type	Input specification for atomic swaps across token types.
DefaultConfiguration	type	Intersection of all sub-wallet and service configurations.

Note: `TokenKind` (`'dust'` | `'shielded'` | `'unshielded'`) and `TokenKindsToBalance` are used internally but are **not exported**.

The facade exposes its sub-wallets as public readonly fields: `shielded`, `unshielded`, `dust`, `submissionService`, `pendingTransactionsService`, `provingService`.

No evidence collected yet.

8. Runtime (@midnight-ntwrk/wallet-sdk-runtime)

Wallet builder and lifecycle management. The runtime infrastructure enables seamless hard-fork migration by dispatching wallet operations to the correct variant based on the current protocol version.

Export	Kind	Description
WalletBuilder	class	<code>init()</code> creates a builder. <code>withVariant(sinceVersion, variantBuilder)</code> registers a variant. <code>build(config?)</code> produces a wallet class.
Runtime	namespace	<code>initHead(args)</code> starts the runtime with the first variant. <code>init(args)</code> starts with a specific variant. <code>dispatch(runtime, impl)</code> invokes a polymorphic operation on the active variant.

Sub-export `./abstractions` :

Export	Kind	Description
Variant	namespace	Variant type definitions and type-level utilities.
VariantBuilder	namespace	Builder types for constructing variants.
WalletLike	namespace	Base wallet interface that all variants implement.
WalletRuntimeError	class	Tagged error (<code>_tag: 'WalletRuntimeError'</code>) with <code>{ message, cause? }</code> .
StateChange	namespace	State transition types for version migration.
VersionChangeType	namespace	Protocol version change classification.

No evidence collected yet.

Appendix A: Generated GraphQL Types (`indexer-client`)

The `indexer-client` package exports ~120 auto-generated types from the indexer's GraphQL schema. These are listed here for completeness.

Symbol	Kind
Connect	const
Disconnect	const
BlockHash	const
FetchTermsAndConditions	const
TransactionStatus	const
DustLedgerEvents	const
ShieldedTransactions	const
UnshieldedTransactions	const
ZswapEvents	const
Maybe	type
InputMaybe	type
Exact	type
MakeOptional	type
MakeMaybe	type
MakeEmpty	type
Incremental	type
Scalars	type
Block	type
BlockOffset	type
CollapsedMerkleTree	type
CommitteeMember	type
ContractAction	type
ContractActionOffset	type
ContractBalance	type
ContractCall	type
ContractDeploy	type
ContractUpdate	type

Symbol	Kind
DParameter	type
DParameterChange	type
DustGenerationDtimeUpdate	type
DustGenerationStatus	type
DustInitialUtxo	type
DustLedgerEvent	type
DustOutput	type
DustSpendProcessed	type
EpochInfo	type
EpochPerf	type
FirstValidEpoch	type
Mutation	type
MutationConnectArgs	type
MutationDisconnectArgs	type
ParamChange	type
PoolMetadata	type
PresenceEvent	type
Query	type
QueryBlockArgs	type
QueryCommitteeArgs	type
QueryContractActionArgs	type
QueryDustGenerationStatusArgs	type
QueryEpochPerformanceArgs	type
QueryEpochUtilizationArgs	type
QueryPoolMetadataArgs	type
QueryPoolMetadataListArgs	type
QueryRegisteredFirstValidEpochsArgs	type

Symbol	Kind
QueryRegisteredPresenceArgs	type
QueryRegisteredSpoSeriesArgs	type
QueryRegisteredTotalsSeriesArgs	type
QuerySpoByPoolIdArgs	type
QuerySpoCompositeByPoolIdArgs	type
QuerySpoIdentitiesArgs	type
QuerySpoIdentityByPoolIdArgs	type
QuerySpoListArgs	type
QuerySpoPerformanceBySpoSkArgs	type
QuerySpoPerformanceLatestArgs	type
QueryStakeDistributionArgs	type
QueryStakePoolOperatorsArgs	type
QueryTransactionsArgs	type
RegisteredStat	type
RegisteredTotals	type
RegularTransaction	type
RelevantTransaction	type
Segment	type
ShieldedTransactionsEvent	type
ShieldedTransactionsProgress	type
Spo	type
SpoComposite	type
SpoIdentity	type
StakeShare	type
Subscription	type
SubscriptionBlocksArgs	type
SubscriptionContractActionsArgs	type

Symbol	Kind
SubscriptionDustLedgerEventsArgs	type
SubscriptionShieldedTransactionsArgs	type
SubscriptionUnshieldedTransactionsArgs	type
SubscriptionZswapLedgerEventsArgs	type
SystemParameters	type
SystemTransaction	type
TermsAndConditions	type
TermsAndConditionsChange	type
Transaction	type
TransactionFees	type
TransactionOffset	type
TransactionResult	type
TransactionResultStatus	type
UnshieldedTransaction	type
UnshieldedTransactionsEvent	type
UnshieldedTransactionsProgress	type
UnshieldedUtxo	type
ZswapLedgerEvent	type
BlockHashQueryVariables	type
BlockHashQuery	type
ConnectMutationVariables	type
ConnectMutation	type
DisconnectMutationVariables	type
DisconnectMutation	type
FetchTermsAndConditionsQueryVariables	type
FetchTermsAndConditionsQuery	type
TransactionStatusQueryVariables	type

Symbol	Kind
TransactionStatusQuery	type
DustLedgerEventsSubscriptionVariables	type
DustLedgerEventsSubscription	type
ShieldedTransactionsSubscriptionVariables	type
ShieldedTransactionsSubscription	type
UnshieldedTransactionsSubscriptionVariables	type
UnshieldedTransactionsSubscription	type
ZswapEventsSubscriptionVariables	type
ZswapEventsSubscription	type
BlockHashDocument	const
ConnectDocument	const
DisconnectDocument	const
FetchTermsAndConditionsDocument	const
TransactionStatusDocument	const
DustLedgerEventsDocument	const
ShieldedTransactionsDocument	const
UnshieldedTransactionsDocument	const
ZswapEventsDocument	const
Query	namespace
Subscription	namespace
QueryClient	class
HttpQueryClient	namespace
SubscriptionClient	class
WsSubscriptionClient	namespace
ConnectionHelper	namespace
QueryRunner	namespace

Appendix B: Evidence Index

Complete index of all verification evidence, organized by test tier and package.

devnet

Test	Tier	Status
devnet/error	tier3	error

standalone

Test	Tier	Status
abstractions/WalletSeed-from-bytes	tier1	pass
abstractions/WalletSeed-from-string	tier1	pass
abstractions/WalletSeed-is-predicate	tier1	pass
abstractions/WalletState-from-string	tier1	pass
abstractions/SerializedTransaction-of	tier1	pass
abstractions/ProtocolVersion-construction	tier1	pass
abstractions/ProtocolVersion-min-max	tier1	pass
abstractions/ProtocolVersion-range-within	tier1	pass
abstractions/ProtocolVersion-is-predicate	tier1	pass
abstractions/NetworkId-well-known	tier1	pass
abstractions/SyncProgress-create	tier1	pass
abstractions/SyncProgress-strictly-complete	tier1	pass
abstractions/SyncProgress-not-complete	tier1	pass
address-format/BLSScalar-is-32-bytes	tier1	pass
address-format/ScaleBigInt-encode-decode-roundtrip	tier1	pass
address-format/ScaleBigInt-zero	tier1	pass
address-format/ScaleBigInt-large-value	tier1	pass
address-format/ShieldedCoinPublicKey-construction	tier1	pass
address-format/ShieldedCoinPublicKey-equality	tier1	pass
address-format/ShieldedEncryptionPublicKey-construction	tier1	pass
address-format/ShieldedAddress-composition	tier1	pass
address-format/UnshieldedAddress-hex-roundtrip	tier1	pass
address-format/UnshieldedAddress-equality	tier1	pass
address-format/DustAddress-from-bigint	tier1	pass
address-format/DustAddress-rejects-over-modulus	tier1	pass
address-format/DustAddress-validates-modulus	tier1	pass
address-format/MidnightBech32m-encode-shielded	tier1	pass

Test	Tier	Status
address-format/MidnightBech32m-encode-unshielded	tier1	pass
address-format/MidnightBech32m-roundtrip	tier1	pass
capabilities/Imbalances-empty	tier1	pass
capabilities/Imbalances-fromEntry	tier1	pass
capabilities/Imbalances-merge	tier1	pass
capabilities/Imbalances-fromEntries	tier1	pass
capabilities/Imbalances-typeSet	tier1	pass
capabilities/InsufficientFundsError-construction	tier1	pass
capabilities/chooseCoin-selects-smallest	tier1	pass
capabilities/chooseCoin-returns-undefined-for-wrong-type	tier1	pass
capabilities/getBalanceRecipe-basic	tier1	pass
capabilities/SubmissionError-construction	tier1	pass
capabilities/SubmissionError-with-cause	tier1	pass
capabilities/ProvingError-construction	tier1	pass
capabilities/ProvingError-tag	tier1	pass
capabilities/PendingTransactions-namespace-exists	tier1	pass
hd/mnemonic-generates-24-words	tier1	pass
hd/mnemonic-validates-roundtrip	tier1	pass
hd/invalid-mnemonic-rejected	tier1	pass
hd/random-seed-length	tier1	pass
hd/wallet-from-seed	tier1	pass
hd/roles-values	tier1	pass
hd/derive-key-at-index-0	tier1	pass
hd/different-roles-different-keys	tier1	pass
hd/different-accounts-different-keys	tier1	pass
hd/composite-role-keys	tier1	pass
hd/deterministic-derivation	tier1	pass

Test	Tier	Status
indexer-client/Connect-query-exists	tier1	pass
indexer-client/Disconnect-query-exists	tier1	pass
indexer-client/BlockHash-query-exists	tier1	pass
indexer-client/TransactionStatus-query-exists	tier1	pass
indexer-client/QueryClient-is-context-tag	tier1	pass
indexer-client/SubscriptionClient-is-context-tag	tier1	pass
indexer-client/ConnectionHelper-deriveWebSocketUrl	tier1	pass
indexer-client/QueryRunner-runPromise	tier1	pass
utilities/SafeBigInt-namespace-exists	tier1	pass
utilities/DateOps-namespace-exists	tier1	pass
utilities/ArrayOps-namespace-exists	tier1	pass
utilities/BlobOps-namespace-exists	tier1	pass
utilities/InvalidProtocolSchemeError-construction	tier1	pass
utilities/FailedToDeriveWebSocketUrlError-construction	tier1	pass
utilities/ClientError-construction	tier1	pass
utilities/ServerError-construction	tier1	pass
wallets/DustWallet-is-factory-function	tier1	pass
wallets/ShieldedWallet-is-factory-function	tier1	pass
wallets/UnshieldedWallet-is-factory-function	tier1	pass
wallets/DustWalletState-has-mapState	tier1	pass
wallets/ShieldedWalletState-has-mapState	tier1	pass
wallets/UnshieldedWalletState-has-mapState	tier1	pass
wallets/createKeystore-from-random-key	tier1	pass
wallets/PublicKey-fromKeyStore	tier1	pass
wallets/WalletBuilder-init	tier1	pass
wallets/Runtime-namespace-exists	tier1	pass
wallets/WalletRuntimeError-construction	tier1	pass

Test	Tier	Status
wallets/FacadeState-exists	tier1	pass
wallets/BalancingRecipe-isRecipe	tier1	pass
wallets/BalancingRecipe-getTransactions	tier1	pass
wallets/WalletFacade-init-is-static	tier1	pass

typecheck

Test	Tier	Status
typecheck/abstractions	tier1	pass
typecheck/address-format	tier1	pass
typecheck/capabilities	tier1	pass
typecheck/dust-wallet	tier1	pass
typecheck/facade	tier1	pass
typecheck/hd	tier1	pass
typecheck/node-client	tier1	pass
typecheck/prover-client	tier1	pass
typecheck/runtime	tier1	pass
typecheck/utilities	tier1	pass

Appendix C: Version Information

Component	Version
Wallet SDK upstream	a1da64624b6a49aeed66d4f6466dee51fbab98ef
Node	0.22.0-rc.6
Indexer	4.0.0-rc.5
Proof Server	8.0.2
Ledger	v8
TypeScript	5.9.3
Node.js	22

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